

Ruqyyah Hassan

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Education

University of Ottawa – Bachelor of Applied Science in Software Engineering (CO-OP) (French Immersion Stream)
2023 – Present, Expected graduation date: 2028

Work Experience

Junior Operations Officer – Statistics Canada (Co-op)

Jan. 2025- Apr. 2025

- Built a Python-based application with a Tkinter GUI to automate quality assurance checks on electronic questionnaire JSON files, reducing manual review.
- Implemented validation logic using regular expressions and added export functionality for Word and Excel reports.
- Enabled user-approved JSON edits by integrating issue previews and auto-fix features directly into the tool.
- Applied object-oriented design and used Git for version control while collaborating with programmers and modernization teams.

Projects

E-commerce sales data analysis – Python

November 2024

- Conducted an analysis of e-commerce sales data to identify key trends and insights in customer purchasing behavior, popular product categories, and seasonal sales fluctuations.
- Used **Pandas**, and **NumPy** functions to manage numerical data, and generate sales summaries based on product categories, customer segments, and regions. Applied **SQL** queries to filter data by product category, purchase date, and customer segment.
- Used **time series analysis** techniques to track seasonal trends and identify peak sales periods. Then I analyzed customer purchasing behavior, including average purchase frequency, basket size, and total spend.
- Visualized sales by category, customer demographics, and monthly trends using **Power BI**.
- Implemented **KMeans clustering** to segment customers and **Linear Regression** for sales forecasting,

Spam mail detection system – Python

October 2024

- Spam emails are a common issue for individuals and filtering them can improve productivity and security.
- My goal was to create a system that could classify emails as either 'spam' or not with high accuracy. The system would need to preprocess raw data and make predictions using machine learning models.
- I used the **scikit** and **panda** libraries. I worked with a data set of labeled emails and started by cleaning the data. Then I transformed the text data into numerical features using the **countVectorizer** classifier. I trained a **random forest classifier** that works well with text classification.
- After training, I evaluated its performance and got an accuracy of 96%.

Job listings website- React

October 2024

- I built a responsive and user-friendly interface that could filter and display job postings efficiently.
- I set up the project using **React**, focusing on functional components, hooks, and responsive design. I implemented a modular design with **useState** and **useEffect** hooks for handling job data dynamically, allowing real-time updates and search functionality. I also ensured mobile responsiveness and used **CSS** for smooth UI performance.

Connect Four game - Java

August 2024

- Developed an arcade version of the Connect Four game using **Java** where two players compete to connect four consecutive blocks of the same color.
- Applied **object-oriented programming (OOP)** by designing classes for players, the game board, and game logic with implementing win conditions and user interactions.

BudgetBuddy – HTML, CSS and Javascript

December 2023

- Collaborated with my team to create a website for a budgeting app that would allow students to manage finances in a group setting.
- Created a Contact Us page to enable the receipt of feedback from users by creating a function using **JavaScript** that sends feedback to our team by email to gather user improvement data.
- Designed the page layout using **HTML** to structure content and **CSS** to style the interface with a focus on user experience to ensure it is appealing and intuitive for the target audience

Portfolio Website – HTML, CSS, jQuery, Bootstrap

December 2023

- Created a personal portfolio website using **HTML, CSS, and Bootstrap** for responsive design, ensuring compatibility across devices.
- Implemented dynamic elements by creating scroll buttons using **jQuery** to allow users to navigate seamlessly between the page's sections.

Skills

Technical Skills:

- Experience in multiple programming languages – **Java, C++, HTML, JavaScript, CSS, Python and Java.**
- Experienced in web development, demonstrated through the completion of my projects.
- Applied attention to detail in coding and website development.

Organizational Skills:

- Successfully planned and executed the BudgetBuddy project, involving coordinating tasks, timelines, and resources with fellow software engineering students.

Communication and Collaboration Skills:

- Collaborated with team members to troubleshoot issues, fostering a positive and cohesive group dynamic.
- Effectively handled and organized payments with extreme attention to detail, ensuring accuracy and transparency in financial transactions.
- Fluency in English, Arabic and French